

$$\begin{aligned}
M_H^2 \rightarrow & -C_H^2 + \frac{1}{16\pi^2} \left(-\frac{1}{4} C_H^2 (g_1^2 + 3g_2^2) - C_H^2 (c_Y \bar{a}_e^{Pr} - s_Y \tilde{\mu} \bar{y}_e^{Pr}) (c_Y a_e^{Pr} - s_Y \tilde{\mu} y_e^{Pr}) \text{LF}_{2,1,0} [m_l^P, m_e^r] + \right. \\
& C_H^2 (c_Y \bar{a}_e^{Pr} - s_Y \tilde{\mu} \bar{y}_e^{Pr}) (c_Y a_e^{Pr} - s_Y \tilde{\mu} y_e^{Pr}) \text{LF}_{3,1,-1} [m_l^P, m_e^r] - \\
& C_H^2 (c_Y \bar{a}_e^{Pr} - s_Y \tilde{\mu} \bar{y}_e^{Pr}) (c_Y a_e^{Pr} - s_Y \tilde{\mu} y_e^{Pr}) \text{LF}_{3,1,0} [m_l^P, m_e^r] + \\
& 3 C_H^2 (c_Y \bar{a}_e^{Pr} - s_Y \tilde{\mu} \bar{y}_e^{Pr}) (c_Y a_e^{Pr} - s_Y \tilde{\mu} y_e^{Pr}) \text{LF}_{4,1,-1} [m_l^P, m_e^r] - \\
& 2 C_H^2 (c_Y \bar{a}_e^{Pr} - s_Y \tilde{\mu} \bar{y}_e^{Pr}) (c_Y a_e^{Pr} - s_Y \tilde{\mu} y_e^{Pr}) \text{LF}_{5,1,-2} [m_l^P, m_e^r] - \\
& 3 C_H^2 (c_Y \bar{a}_d^{Pr} - s_Y \tilde{\mu} \bar{y}_d^{Pr}) (c_Y a_d^{Pr} - s_Y \tilde{\mu} y_d^{Pr}) \text{LF}_{2,1,0} [m_q^P, m_d^r] + \\
& 3 C_H^2 (c_Y \bar{a}_d^{Pr} - s_Y \tilde{\mu} \bar{y}_d^{Pr}) (c_Y a_d^{Pr} - s_Y \tilde{\mu} y_d^{Pr}) \text{LF}_{3,1,-1} [m_q^P, m_d^r] - \\
& 3 C_H^2 (c_Y \bar{a}_d^{Pr} - s_Y \tilde{\mu} \bar{y}_d^{Pr}) (c_Y a_d^{Pr} - s_Y \tilde{\mu} y_d^{Pr}) \text{LF}_{3,1,0} [m_q^P, m_d^r] + \\
& 9 C_H^2 (c_Y \bar{a}_d^{Pr} - s_Y \tilde{\mu} \bar{y}_d^{Pr}) (c_Y a_d^{Pr} - s_Y \tilde{\mu} y_d^{Pr}) \text{LF}_{4,1,-1} [m_q^P, m_d^r] - \\
& 6 C_H^2 (c_Y \bar{a}_d^{Pr} - s_Y \tilde{\mu} \bar{y}_d^{Pr}) (c_Y a_d^{Pr} - s_Y \tilde{\mu} y_d^{Pr}) \text{LF}_{5,1,-2} [m_q^P, m_d^r] - \\
& 3 C_H^2 (s_Y \bar{a}_u^{Pr} - \tilde{\mu} c_Y \bar{y}_u^{Pr}) (s_Y a_u^{Pr} - \tilde{\mu} c_Y y_u^{Pr}) \text{LF}_{2,1,0} [m_u^r, m_q^P] + \\
& 3 C_H^2 (s_Y \bar{a}_u^{Pr} - \tilde{\mu} c_Y \bar{y}_u^{Pr}) (s_Y a_u^{Pr} - \tilde{\mu} c_Y y_u^{Pr}) \text{LF}_{3,1,-1} [m_u^r, m_q^P] - \\
& 3 C_H^2 (s_Y \bar{a}_u^{Pr} - \tilde{\mu} c_Y \bar{y}_u^{Pr}) (s_Y a_u^{Pr} - \tilde{\mu} c_Y y_u^{Pr}) \text{LF}_{3,1,0} [m_u^r, m_q^P] + \\
& 9 C_H^2 (s_Y \bar{a}_u^{Pr} - \tilde{\mu} c_Y \bar{y}_u^{Pr}) (s_Y a_u^{Pr} - \tilde{\mu} c_Y y_u^{Pr}) \text{LF}_{4,1,-1} [m_u^r, m_q^P] - \\
& 6 C_H^2 (s_Y \bar{a}_u^{Pr} - \tilde{\mu} c_Y \bar{y}_u^{Pr}) (s_Y a_u^{Pr} - \tilde{\mu} c_Y y_u^{Pr}) \text{LF}_{5,1,-2} [m_u^r, m_q^P] + \\
& \frac{3}{2} C_H^2 g_1^2 \text{LF}_{2,1,-1} [\tilde{\mu}, m_1] + 2 m_1 s_Y C_H^2 \tilde{\mu} c_Y g_1^2 \text{LF}_{2,1,0} [\tilde{\mu}, m_1] - C_H^2 g_1^2 \text{LF}_{3,1,-2} [\tilde{\mu}, m_1] + \\
& 2 C_H^2 g_1^2 (C_H^2 - m_1 s_Y \tilde{\mu} c_Y) \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] + 2 m_1 s_Y \tilde{\mu} c_Y g_1^2 C_H^2 \text{LF}_{3,1,0} [\tilde{\mu}, m_1] - \\
& 4 g_1^2 C_H^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] - 6 m_1 s_Y \tilde{\mu} c_Y g_1^2 C_H^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + \\
& 2 g_1^2 C_H^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + 4 m_1 s_Y \tilde{\mu} c_Y g_1^2 C_H^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \\
& \frac{9}{2} C_H^2 g_2^2 \text{LF}_{2,1,-1} [\tilde{\mu}, m_2] + 6 m_2 s_Y C_H^2 \tilde{\mu} c_Y g_2^2 \text{LF}_{2,1,0} [\tilde{\mu}, m_2] - 3 C_H^2 g_2^2 \text{LF}_{3,1,-2} [\tilde{\mu}, m_2] + \\
& 6 C_H^2 g_2^2 (C_H^2 - m_2 s_Y \tilde{\mu} c_Y) \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] + 6 m_2 s_Y \tilde{\mu} c_Y g_2^2 C_H^2 \text{LF}_{3,1,0} [\tilde{\mu}, m_2] - \\
& 12 g_2^2 C_H^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] - 18 m_2 s_Y \tilde{\mu} c_Y g_2^2 C_H^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + \\
& \left. 6 g_2^2 C_H^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] + 12 m_2 s_Y \tilde{\mu} c_Y g_2^2 C_H^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \right)
\end{aligned}$$